1

a

$\hat{M}_{MLE} = x_n$

b

Given from the announcement:

$E(\hat{M}_{MLE}) = \frac{n}{n+1}(M+1)$

> This is a biased estimator, as $\hat{M}_{ME} \leq M$ and will underestimate M.

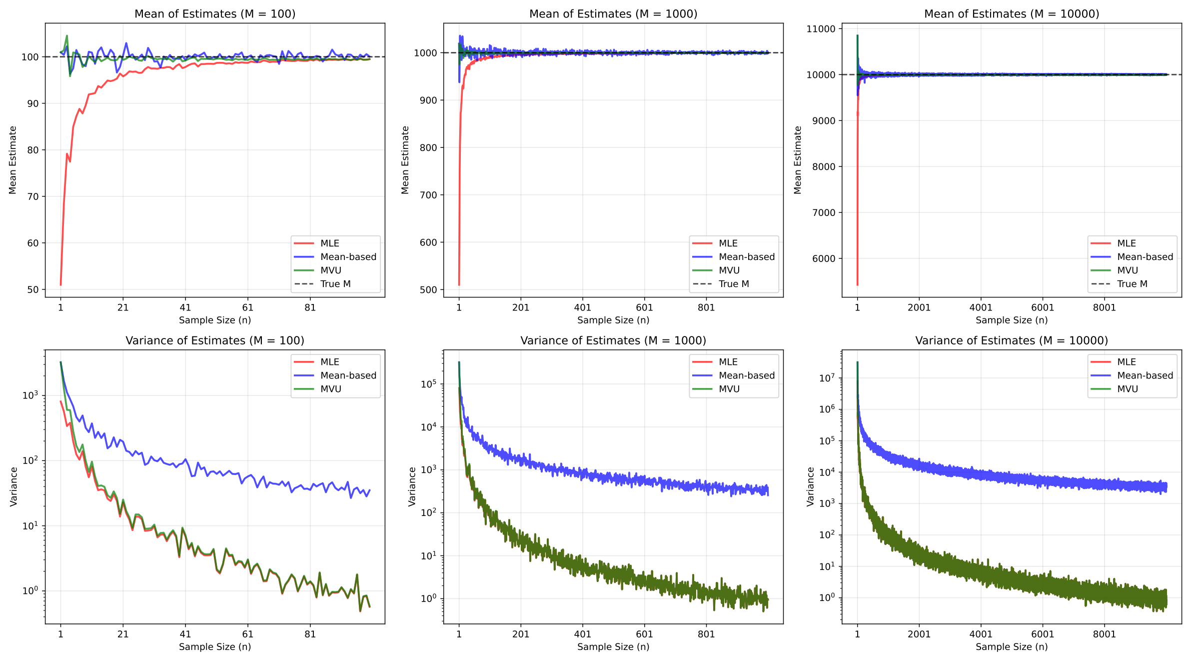
$E(\hat{M}_{MEAN}) = M$

> This is an unbiased estimator as the expected value of M_{MEAN} is equal to the value it is trying to estimate (M)

$E(\hat{M}_{MVU}) = M$

> This is an unbiased estimator as the expected value of M_{MVU} is equal to the value it is trying to estimate (M)

c



According to the Mean of estimates graph it is clear to see that the $\hat{M}_{MEAN}$ and $\hat{M}_{MVU}$ estimates are unbiased estimators of M as they their expected value is M. On the other hand, $\hat{M}_{MLE}$ is evidently an unbiased estimator as it falls significantly below M for smaller values of n. For sufficiently large n, the three estimators all approach M which is good.

Their variances on the other hand tell a different story. The $\hat{M}_{MEAN}$ estimator has an extremely volatile variance many orders of magnitude above both $\hat{M}_{MLE}$ and $\hat{M}_{MVU}$. This demonstrates that the $\hat{M}_{MVU}$ estimator should be the best estimator for M as the variance is tied for smallest and is an unbiased estimator of M.

2

a

b

c

d

3

a

The number of US cars is 249 and the number of Japanese cars is 79 in this dataset.

b

Let μ_{US} be the population mean mileage for US cars and μ_{JP} be the population mean mileage for Japanese cars.

$H_0 : \mu_{US} = \mu_{JP}$ and $H_1 : \mu_{US} < \mu_{JP}$

c

To test our hypothesis we can use a two sample t-test. The test statistic is:

$$t = \frac{\bar{x}_{JP} - \bar{x}_{US}}{\sqrt{\frac{s_{JP}^2}{n_{JP}} + \frac{s_{US}^2}{n_{US}}}}$$

But first we would check the normaility and the variances of the two car samples to see if this is the right test. When checking, the normaility, we find that the US cars are not normally distributed. However using the central limit theorem (both samples are larger than 30), we can say that the sample mean approaches a normal distribution so we will continue with our tests.

d

The value of the test statistic is 12.946, resulting in a p-value of 7.854×10^{-26}.

e

Since the p-value is smaller than any reasonable α we reject the null hypothesis and there is enough statistical evidence to support the salesman's claim.